Xiaotian Zhou

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Education

The Pennsylvania State University 08/2025–Present
Ph.D. Candidate, Computer Science and Engineering University Park, PA
Advisors, Prof. Syed Rafiul Hussain & Prof. G. Gary Tan

The Pennsylvania State University 08/2023–05/2025
Master of Science, Computer Science and Engineering University Park, PA
Advisor, Prof. Syed Rafiul Hussain

ShanghaiTech University 08/2019–06/2023
Bachelor of Engineering, Computer Science and Technology Shanghai, China
Visiting Student, University of California, Berkeley (08/2021–05/2022)
Thesis, Mitigating Adversarial Attacks against Synthetic Audio Detection Systems

Research Interests

Formal Verification, Computer Security, Programming Languages, Proof Assistants, Cryptography

Research Experience

The Pennsylvania State University 09/2023–Present
Research Assistant University Park, PA

- **Verified ASN.1 UPER Parser**: Built a generator of verified UPER parsers in Rocq with bit-level round-trip proofs, and used it as a differential-testing oracle to find bugs in 11 widely used ASN.1 parsers [2].
- **Rust Security**: Designing and implementing a sanitizer for Rust aliasing undefined behavior.

Seagate Technology 05/2026–08/2026
Research Intern - Security and Cryptography Shakopee, MN

- Developed a computational security proof of HPKE (RFC 9180) as used in OCP L.O.C.K., the Open Compute Project's key-management specification for self-encrypting drives, using SSProve and Hax.

Publications

[1] Kai Tu, Tianchang Yang, Xiaotian Zhou, Ali Ranjbar, Abdullah Al Ishtiaq, Tianwei Wu, Yilu Dong, and Syed Rafiul Hussain. **Octopus: Target-Aware Adaptive Field-level Fuzzing for Cellular ASN.1 Protocols**. *To appear in* IEEE Symposium on Security and Privacy (S&P), 2027.

[2] Xiaotian Zhou, Kai Tu, Ali Ranjbar, Yilu Dong, Gang Tan, and Syed Rafiul Hussain. **VUPER: Verified ASN.1 UPER Parser**. The ACM Conference on Computer and Communications Security (CCS), 2026.

Preprints

[P1] Nan An and Xiaotian Zhou. **A Simplified Layout Algebra for GPU Kernels**. *arXiv preprint* arXiv:2610.13679, 2026.

Teaching Experience

The Pennsylvania State University 08/2025–12/2025
Teaching Assistant - Computer Security University Park, PA
- Designed and graded programming assignments and quiz questions for a class of 70+ students.
- Held weekly office hours to guide students on Cryptography and Computer Security concepts.

ShanghaiTech University 09/2020–06/2021, 01/2023–06/2023
Teaching Assistant - Mathematical Analysis I & II Shanghai, China
- Conducted weekly recitation sessions for 30+ students.
- Graded assignments and exams on topics like Calculus.

Professional Service

Artifact Evaluation Committee: CCS '26
External Reviewer: FutureG '26

Awards

- OPLSS Fellowship ($1,780) *Oregon Programming Languages Summer School* 2024
- Exchange Student Program Scholarship *ShanghaiTech University* 2022
- Merit Student *ShanghaiTech University* 2019–2020

Security Findings

- **Unisoc**: Discovered 8 High-severity CVEs with Octopus [1], such as CVE-2025-71256, CVE-2026-21549, and CVE-2026-21550.
- **Apple**: Discovered a baseband vulnerability with Octopus [1], acknowledged in iOS 27 security notes.
- **asn1c**: Discovered CVE-2025-55398 (Critical) with VUPER [2].
- **Rocq Prover**: Discovered a critical soundness bug permitting a proof of `False`.

Skills

- **Relevant Coursework:** Computer Security, Cryptography, Parallel Programs & Processors, Machine Learning, Operating Systems, Compiler Construction, Securing Embedded Systems
- **Programming Languages:** C, C++, Python, Rocq Prover (Coq), OCaml, Rust
- **Software:** Docker, Git, LaTeX, Linux, PyTorch, NumPy, Ghidra, AFL++, LibAFL